CEIUPM | Centro de Electrónica Industrial

Wireless Sensor Networks: Lab 1 - Exercises 29/09/2025

For the following exercises edit the examples used during the practical lab related to the use of LEDs, UDP client and server transmissions, etc.

Based on that, proceed with the following exercises:

1. According to the lab example in which 2 nodes sent packets, toggle the status of a **LED** (ON/OFF) whenever a new packet is received. Show how LEDs are toggled in any of the two visualization options in the Cooja simulator.

2. Use the example of the RPL-UDP network, configure the simulation with 7 nodes (one of them as a receiver and the rest as senders) and send periodically (every 2 seconds) a simulated value of a **temperature sensor** from the sensor nodes to the server (you can use, for instance, the standard *sprintf* or *snprintf* functions to save the string in a buffer, and then send it with the available functions in the sender code). To simulate the temperature values, you can use a random number between coherent values of ambient temperature (the function random_rand can help you with it). Show how the RPL routes are created and how the effect of changing the position of a node will modify the path from the sender to the receiver.

Evaluation

The exercises will be evaluated through a document that should include the following items:

- Code of each exercise (inside the document).
- Brief explanation with screen captures showing the results of the exercise.
- Upload this document to **Moodle** by Monday the **7th of October** (a specific homework task will be created in Moodle for it).
- Remember this is part of the evaluation score of the subject.